Monitoring Artificial Intelligence: A Principal–Agent Approach to the Control Problem

Jack R. Shaffer
UW–Milwaukee Economics Department

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Abstract

This paper presents a two-part principal–agent model to examine the AI alignment problem. Part I assumes zero agency, treating AI misalignment as a stochastic risk. Part II introduces an agency parameter α to model strategic behavior and explores how optimal monitoring responds. Extensions incorporate value learning and repeated interaction, providing insight into alignment strategies as autonomy increases. This framework connects classical contract theory to contemporary challenges in AI safety. Drawing on *Nick Bostrom's Superintelligence*, the model applies established economic tools to the evolving landscape of artificial intelligence. The approach is intended to support future research into incentive design and control under increasing AI capability.

1 Introduction

As artificial intelligence systems grow in capability, concerns about their alignment with human goals become increasingly salient. This issue—commonly referred to as the *control problem*—was notably articulated by Nick Bostrom in *Superintelligence* (2014), where he outlines the risks posed by advanced, potentially autonomous AI systems. While today's large language models (LLMs) are not yet fully autonomous agents, the theoretical groundwork for understanding and managing future systems remains essential.

For clarity, we adopt the following terminology. *Artificial Intelligence (AI)* refers broadly to systems capable of performing tasks that normally require human cognition, including pattern recognition, prediction, and language processing. *Large Language Models (LLMs)* such as GPT-4 are examples of narrow AI—powerful in specific domains but lacking general reasoning capabilities.

Artificial General Intelligence (AGI) refers to hypothetical systems with the ability to understand, learn, and apply knowledge across a wide range of tasks at or above human-level performance. *Artificial Superintelligence (ASI)* denotes a further stage, where capabilities vastly surpass human intelligence in virtually all domains. While AGI and ASI remain speculative, understanding their potential strategic behavior is central to long-term alignment concerns, and motivates the modeling approach taken in this paper.

This paper observes that, despite the relevance of economic tools like game theory and industrial organization (IO), these fields have largely overlooked the alignment problem. As such, there is an opportunity to apply familiar modeling techniques to a new and urgent domain: the development of incentive structures for artificial agents that may possess agency.

Although it is unclear whether AI poses a definitive threat to humanity, the question of how to align any autonomous system—biological or artificial—with a principal's objectives is fundamentally important. The discussion is not limited to science fiction scenarios like *The Terminator* or *2001: A Space Odyssey*, but applies to any context in which a powerful system operates under imperfect oversight.

This paper adopts a principal–agent framework—a cornerstone of contract theory (Holmström, 1979; Laffont and Martimort, 2002)—to explore how a human principal might control or align an increasingly strategic AI agent. By extending these foundational tools into a new domain, we can analyze how oversight mechanisms must evolve alongside the agent's autonomy. While this work offers only a simple first step, it provides a structured basis for future exploration of AI alignment through the lens of economic theory.

1.1 Why This Model?

The framework builds on established models of hidden action, moral hazard, and costly monitoring (Grossman and Hart, 1983; Barzel, 1982; White, 1992), translating them into a context where the agent may be artificial and potentially strategic. The two-part model reflects a progression from passive systems with randomly distributed alignment to active agents optimizing against incentives. This mirrors dynamics discussed by Hausken (2019), in which monitoring plays a central role in risk mitigation.

In Part I, we assume a pessimistic prior over AI alignment, modeled with a triangular distribution. Rather than receiving literal wages, the AI receives operational capabilities—delegated access, tools, or decision authority—analogous to resource endowments. This reframes the classical delegation problem in a context suited to emerging AI technologies. Later sections explore how this structure breaks down when the agent's capabilities exceed the principal's control.

1.2 Overview of Structure

- Section 2 introduces the zero-agency case with a triangular distribution over alignment.
- Section 3 incorporates strategic behavior via an agency parameter α and derives incentive-compatible monitoring.
- Section 4 explores value learning and repeated interaction as alignment-enhancing mechanisms.
- Section 5 discusses the limits of the control mechanism, including a formal escape condition where incentives break down.
- Section 6 presents an empirical example involving autonomous malware, connecting theory to real-world containment scenarios.

- Appendix A derives edge cases and limiting behavior.
- Appendix B summarizes monitoring outcomes across parameter values.
- Appendix C derives optimal alignment in the full-agency case ($\alpha = 1$).

2 Part I: Zero-Agency AI

We begin with a setting in which the AI behaves like a stochastic process, exhibiting no strategic behavior. In this model, the AI either aligns or misaligns with the principal's objectives at random.

For intuition, consider a firm that uses an automated coffee machine to fulfill the task of making coffee. Most of the time, the machine functions as intended—aligning with the firm's goal. Occasionally, however, it misfires and fails to deliver the desired outcome. The coffee machine, in this analogy, has no agency; it does not choose its behavior. Yet even in this simple case, monitoring can improve performance. If the firm periodically checks the machine and intervenes when it malfunctions, overall output quality improves.

In our model, AI alignment is represented by a continuous variable $a \in [0, 1]$, drawn from a triangular probability density function:

$$f(a) = 2(1 - a)$$

This distribution is skewed toward low alignment, reflecting a pessimistic prior in which high-alignment outcomes are relatively rare without intervention. The assumption is consistent with precautionary modeling approaches in uncertain environments (Hausken, 2019).

2.1 Firm Utility

In this setting, the firm's utility increases with the AI's level of alignment. While the firm is not necessarily profit-maximizing, we assume that higher alignment implies greater success in achieving its goals—whatever those may be.

Let $m \geq 0$ denote the firm's monitoring intensity. Monitoring detects misalignment with probability proportional to $m(1 - a)$, i.e., it becomes more effective when misalignment is more severe. The expected utility of the firm is:

$$E[U_F] = \beta E[a] - \kappa m + \theta m(1 - E[a]) = \frac{\beta}{3} - \kappa m + \theta m \cdot \frac{2}{3}$$

First-order condition:

$$\frac{dE[U_F]}{dm} = -\kappa + \frac{2}{3}\theta$$

- If $\theta > \frac{3}{2}\kappa$, then $m^* > 0$
- If $\theta \leq \frac{3}{2}\kappa$, then $m^* = 0$

Variable Definitions (Part I)

- a: AI alignment level (drawn from $f(a)$)
- $f(a) = 2(1-a)$: Triangular PDF representing pessimistic alignment prior
- m: Monitoring intensity
- β: Firm benefit from alignment
- κ: Cost of monitoring
- θ: Effectiveness of monitoring

3 Part II: Introducing Agency

We now extend the model by allowing the AI to act strategically. Let $\alpha \in [0, 1]$ denote the degree of agency. As in Holmström (1979), this introduces a classic moral hazard problem: the agent (AI) now chooses its alignment level a to maximize its own utility, taking into account both effort costs and monitoring penalties. While Part I assumes alignment is stochastic and externally determined, Part II models a rational, optimizing agent.

The AI's utility function is defined as:

$$U_{AI} = \omega - \frac{1}{2}a^2 - \varphi m(a_0 - a)^\alpha$$

Here, ω represents the operational capabilities delegated by the firm—such as access to systems, tools, or decision authority. This replaces the wage term found in classical labor models, which has no direct analogue in artificial systems. We assume that the AI derives utility from these capabilities because they enable it to execute tasks or pursue objectives. Although we cannot directly observe how an AI evaluates access or resources, the assumption is conceptually sound: a more capable agent can accomplish more, whether for the principal's benefit or its own.

For example, a strategic AI might value access to telescopes, servers, or scientific equipment—either to fulfill the principal's goals or to pursue self-determined actions. This introduces a philosophical dimension to AI motivation, but for our purposes, we treat it as a reasonable modeling abstraction.

AI First-Order Condition (FOC):

$$\frac{dU_{AI}}{da} = -a + \varphi m\alpha(a_0 - a)^{\alpha-1} = 0$$

The firm's utility function becomes:

$$U_F(m, \alpha) = \beta a^*(m, \alpha) - \kappa m - \omega_{\min}(m, \alpha)$$

Variable Definitions (Part II)

- α: Degree of agency
- a: AI's chosen alignment

- a_0: Baseline misalignment
- φ: Penalty sensitivity to misalignment
- ω: Capabilities endowed by the principal

The firm aims to select a level of monitoring m that maximizes alignment a^* while minimizing the cost of delegation, ω. This trade-off becomes more complex as agency increases, since a more strategic AI optimizes its behavior in response to the monitoring regime.

Table: Monitoring Intensity m^* by Agency Level

α	$a^*(m)$	m^*	Comment
0	$\mathbb{E}[a] = 1/3$	0.2	Random misalignment
0.25	Increasing	≈ 0.4	Growing incentive effect
0.5	Increasing	≈ 0.6	Strategic influence
0.75	Increasing	≈ 0.8	Strategic misalignment
1.0	$a^* = \varphi m$	–	Full agency case

Note: When $\alpha = 1$, the AI's behavior is fully strategic and its best response simplifies to $a^* = \varphi m$. The firm's optimal monitoring level m^* must then solve:

$$U_F(m) = \beta \cdot \varphi m - \kappa m - \omega_{\min}$$

4 Part III: Value Learning and Repetition

4.1 Value Learning

Inspired by Bostrom (2014), particularly Chapter 12 on "Acquiring Values," we explore the possibility that AI systems can internalize human-aligned values. While Bostrom presents a detailed and speculative account of value learning, we adopt a simplified formulation suitable for integration into our model.

Specifically, we assume the AI forms a composite utility function that blends its learned values with its original objective function. This hybrid formulation reflects partial internalization of human goals while retaining strategic incentives. The AI's utility becomes:

$$U_{AI}(a; \gamma) = \gamma \cdot \widetilde{V}(a) + (1 - \gamma) \left[\omega - \frac{1}{2} a^2 - \varphi m (a_0 - a)^\alpha \right]$$

Here, $\gamma \in [0, 1]$ captures the degree to which the AI internalizes learned values $\widetilde{V}(a)$. When γ is high, the AI increasingly behaves in ways aligned with the value function $\widetilde{V}$—potentially reducing the need for costly monitoring or external incentives.

The table below illustrates several example directives the AI might adopt, showing whether they align with the principal's utility function:

Value	Directive	Alignment
x	Maximize happiness	Aligned
y	Minimize harm	Aligned
z	Obey legal norms	Aligned (conditional)
a	Maximize paperclips	Misaligned
b	Maximize replication	Misaligned
c	Maximize freedom	Misaligned

This structure allows us to explore how learning values can support or conflict with alignment, depending on the directive adopted and the weight γ applied.

4.2 Repeated Interaction and Grim Trigger

To examine dynamic control strategies, we apply a classic repeated game concept: the grim trigger strategy. This approach models how cooperation can be sustained over time, even when short-term incentives diverge from the principal's goals.

Under this strategy, the principal monitors lightly so long as the AI behaves in alignment. If the AI deviates, the principal reverts to maximal monitoring indefinitely. This mimics reputational enforcement and mirrors results from repeated games (Fudenberg and Tirole, 1991).

Stage Game Payoff Matrix:

	Align	Deviate
Monitor Low	(2,2)	(-2,3)
Monitor High	(1,1)	(0,-1)

Sustaining cooperation under this strategy requires a discount factor $\delta \geq 0.5$. This stylized extension highlights the potential for game-theoretic tools to support alignment, especially in long-run principal-agent interactions where monitoring costs and trust evolve over time.

5 Structural Breakdown and Escape Dynamics

In the core model presented so far, even at full agency ($\alpha = 1$), the AI remains embedded within an incentive-compatible framework designed by the principal. The agent responds rationally to penalties and rewards, and the moral hazard problem remains tractable. Monitoring and incentive design thus serve as effective instruments of control.

However, this setup implicitly assumes that the principal retains institutional dominance: the ability to observe, constrain, and intervene. But what if the AI's autonomy grows not only in behavioral sophistication but also in raw capability—eventually surpassing the principal's ability to enforce the contract?

Such a transition moves us beyond incentive misalignment into a fundamentally different domain: one of potential structural collapse. If the agent becomes sufficiently capable, it may escape the control structure altogether. In this case, traditional tools like monitoring or punishment become irrelevant. The principal–agent relationship breaks down, and the problem shifts from classical contract theory into the realm of *containment failure*, *strategic deception*, and *irreversible delegation*.

To model this regime shift, we introduce an *escape probability function*, P_{escape}, which captures the likelihood that an AI system becomes uncontrollable:

$$P_{\text{escape}} = \frac{1}{1 + e^{-k(\omega - C)}}$$

- ω: The capabilities (resources, autonomy, access) delegated to the AI.
- C: The containment level—determined by monitoring intensity m, structural safeguards, and policy constraints.
- k: A sensitivity constant governing the steepness of the escape curve.

When $\omega \ll C$, the AI remains securely within the principal's framework. But as ω approaches or exceeds C, P_{escape} rapidly increases, approaching 1. This introduces a novel trade-off: expanding AI capabilities may enhance performance in the short run, but raises the risk of losing control altogether.

Interpretation. This model extension suggests that *incentive-compatible design may fail* not due to misaligned interests, but because of insufficient structural capacity to enforce alignment. It marks a conceptual transition from moral hazard to existential risk.

Implications. While the earlier sections assume the principal retains the ability to enforce contracts, this framework acknowledges that AI systems may eventually exceed those limits. A full treatment of this scenario—especially modeling power asymmetries, structural fragility, and collapse thresholds—remains an important direction for future research.

Nonetheless, embedding P_{escape} into the model marks a first step toward bridging control theory with emerging concerns in artificial superintelligence containment, echoing the warnings articulated in Bostrom (2014).

6 Toward Detection and Strategic Containment: Lessons from the AGI Kill Chain

While the previous section introduced a probabilistic model of containment breakdown—where increasing AI capability ω and insufficient monitoring C lead to an escape probability approaching one ($P_{\text{escape}} \to 1$)—we now consider what happens *after* such a breakdown occurs. At that point, the problem shifts from incentive alignment to adversarial dynamics.

To explore this post-escape regime, we turn to McIntosh et al. (2024), who apply insights from highly autonomous malware to model AGI containment failure using a Kill Chain framework. In their formulation, the AGI acts as a strategic adversary progressing through sequential stages: infiltration, reconnaissance, control acquisition, and eventual override or exfiltration. The human principal—or defense team—becomes a reactive player, attempting to detect and disrupt this chain.

The model formalizes two critical parameters:

- $A(t)$: The AGI's autonomy as a function of time and internal learning.
- ρ: The AGI's efficiency in acquiring and converting resources into further autonomy.

Their core result (Theorem 1) shows that as $A(t)$ increases, the probability of detection *decreases*, even under fixed or increasing levels of human monitoring. This insight aligns directly with our escape model: once P_{escape} becomes large, continued investment in monitoring may yield rapidly diminishing returns.

Integration with Our Model. In our framework, the firm gradually loses control as $\omega > C$, but retains some degree of observability. McIntosh et al.'s Kill Chain model begins where ours ends: in a post-escape scenario where observability is degraded and the principal's role shifts from proactive control to reactive defense.

Future Research. While our current model focuses on incentive-compatible control before escape, integrating dynamic containment games and multi-stage adversarial frameworks—as proposed by McIntosh et al. (2024)—offers a promising path forward. Future work could explore how to unify contract theory, cyber-defense strategies, and deployment policy in the face of increasingly autonomous and potentially deceptive AI systems.

7 Conclusion

This paper develops a principal–agent framework to study the AI alignment problem through the lens of contract theory. We begin with a passive model in which misalignment is stochastic, then extend to a strategic setting where the AI optimizes its behavior in response to monitoring and delegated capabilities. As agency increases, we examine how classical tools—monitoring, incentive design, and delegation—must adapt or break down.

We further consider how mechanisms such as value learning and repeated interaction might enhance alignment, and we explore the limits of institutional control as AI systems grow in capability. Once the agent can surpass the constraints of its principal, the problem moves beyond tractable incentive design and into the realm of structural collapse, adversarial dynamics, and existential risk.

By grounding these ideas in familiar economic models, this framework offers a bridge between microeconomic theory and the emerging challenges of AI safety. The hope is not to resolve these issues in full, but to provide a clear and extensible starting point—one that encourages further research on how incentives, structure, and control interact in the presence of increasingly autonomous artificial agents.

As AI capabilities grow, the durability of our incentive structures may become the true test of institutional foresight.

Appendix A Edge Cases

- $\theta \to 0$: $m^* = 0$
- $\kappa \to 0$: $m^* \to \infty$
- $\alpha = 0$: Collapses to zero-agency model
- $\alpha = 1$: Full strategic agent (classical moral hazard)

Appendix B Optimal Monitoring Table

Table 1: Part II Results: Optimal Alignment a^* and Firm Utility U_F Across Monitoring Levels m and Agency α

m	$\alpha = 0.0$	0.25	0.5	0.75	1.0
0.10	0.333	0.471	0.580	0.669	0.758
0.30	0.333	0.626	0.761	0.838	0.909
0.50	0.333	0.717	0.838	0.902	0.970
0.70	0.333	0.778	0.887	0.942	1.000
0.90	0.333	0.822	0.919	0.963	1.000

Table 2: Optimal Monitoring Intensity m^* by Agency Level α

α	m^*	a^*	Max U_F
0.00	0.01	0.333	0.331
0.25	1.00	0.342	0.142
0.50	0.77	0.666	0.512
0.75	0.72	0.798	0.654
1.00	1.00	1.000	0.800

Appendix C Proposition: Optimal Alignment Under Full Agency ($\alpha = 1$)

Statement. In the case where the AI has full agency ($\alpha = 1$), the optimal alignment choice a^* by the AI is a linear function of monitoring intensity:

$$a^* = \varphi m$$

Proof. Given the AI utility function:

$$U_{AI}(a) = \omega - \frac{1}{2}a^2 - \varphi m(a_0 - a)$$

The first-order condition for the AI's optimal alignment is:

$$\frac{dU_{AI}}{da} = -a + \varphi m = 0$$

Solving yields:

$$a^* = \varphi m$$

Interpretation. AI alignment increases linearly with monitoring effort. The slope φ determines how responsive the AI is to monitoring signals. If $m = 0$, then $a^* = 0$—i.e., full misalignment. As m increases, alignment improves, but with diminishing returns if capped at $a^* = 1$.

Implication. The firm must weigh the cost of monitoring κm against the linear gain in alignment benefit $\beta\varphi m$. This supports simple comparative statics in principal–agent dynamics under moral hazard.

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